

Practice Problems – Set 2

Answer Key

This practice is based on Chapters 12–20 of Feenstra & Taylor. Work through all questions to prepare for the Final Exam.

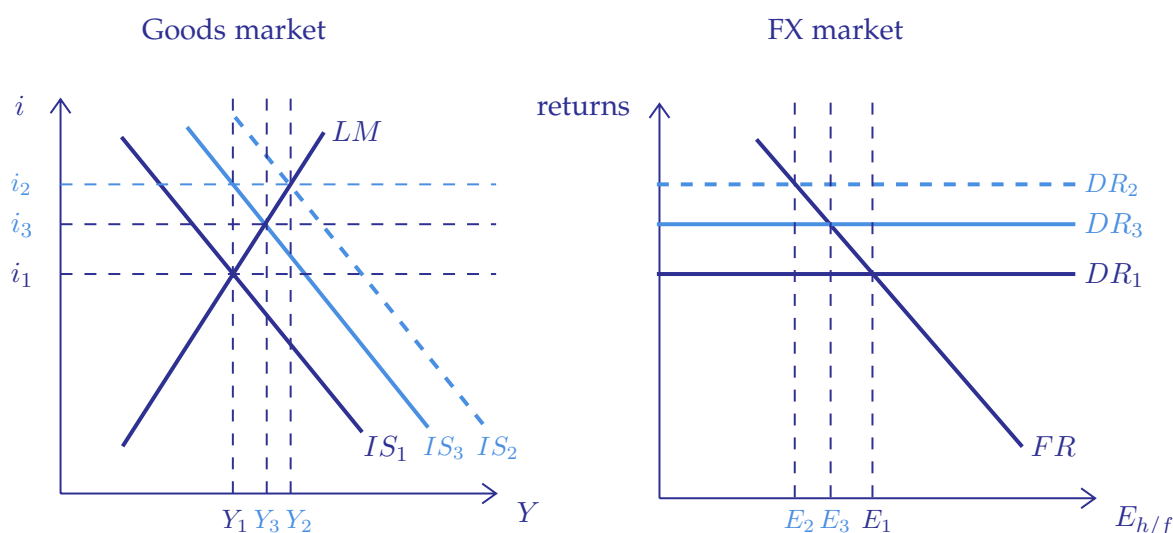
1. (15 points) IS-LM-FX: Fiscal Policy Under Float and Fixed.

Home is a small open economy. Starting from equilibrium (Y_1, i_1, E_1) , the home government permanently increases spending G .

- (a) Trace the full adjustment in the IS-LM-FX model under a **floating** exchange rate. Which curves shift, in what direction, and why? Use a diagram to show the new equilibrium (Y_3, i_3, E_3) .

$G \uparrow$ shifts the **IS right** ($IS_1 \rightarrow IS_2$). Higher income raises money demand, pushing i up along LM. Higher i attracts capital inflows, causing the exchange rate to **appreciate sharply on impact** — overshooting to E_2 , a level more appreciated than the new long-run equilibrium. This large appreciation worsens the trade balance substantially, shifting IS **partially back left** ($IS_2 \rightarrow IS_3$). As E depreciates from its overshooting level back toward E_3 , expected depreciation is positive ($E^e > E_2$), which via UIP allows $i_3 > i^*$ without further capital flows. The economy settles at IS_3/LM , with E at its new long-run level E_3 .

Final equilibrium: $Y_3 > Y_1$ (output rises, but less than under fixed), $i_3 > i_1$ (interest rate rises), $E_1 > E_3 > E_2$ (currency appreciated at long run, but overshoots further on impact). Fiscal policy **partially raises output**: the TB crowding-out is real but incomplete, so output does rise — unlike the static-expectations Mundell-Fleming special case where $i = i^*$ is imposed and crowding out is complete.



IS_2 (dashed): impact shift from $G \uparrow$. IS_3 (solid blue): final position after partial TB crowding-out. E_2 is the overshooting impact exchange rate; E_3 is the long-run equilibrium.

librium ($E_1 > E_3 > E_2$). DR_3 intersects FR at E_3 , consistent with $i_3 > i^*$ satisfying UIP via expected depreciation from E_2 back to E_3 .

- (b) What happens to the home trade balance? Explain intuitively why fiscal policy only *partially* raises output under floating.

The trade balance *worsens*. The currency appreciation (both on impact and at the long-run equilibrium $E_3 < E_1$) makes home goods more expensive for foreigners (exports fall) and foreign goods cheaper for home consumers (imports rise).

Intuition: the appreciation partially crowds out net exports, offsetting some of the demand stimulus from G . But this crowding-out is incomplete: the IS only shifts partway back (IS_3 is still to the right of IS_1), so output does rise — just less than the full multiplier. The key difference from a fixed exchange rate is precisely this TB channel: under floating, the appreciation works against the fiscal expansion; under fixed, it is absent because E cannot move.

- (c) How would your answer to (a) change under a **fixed** exchange rate regime? Name the two key differences in the adjustment mechanism.

Under a fixed exchange rate: (1) The appreciation pressure is eliminated by the central bank, which *expands the money supply* (buys foreign currency) to keep E constant, shifting LM to the right. (2) Because E never changes, there is no TB crowding-out and no IS correction. Output rises by the full multiplier amount, $Y_3^{\text{fixed}} > Y_3^{\text{float}} > Y_1$. Under floating, the exchange rate channel partially undoes the fiscal stimulus; under fixed, that channel is shut down entirely, making fiscal policy more powerful.

2. (12 points) External Wealth and Valuation Effects.

Country Luzia has the following external balance sheet (in billions of local currency, the lúz):

External Assets		External Liabilities	
100% denominated in lúz	400	80% denominated in dollars	600

The current exchange rate is $E = 1$ lúz per dollar.

- (a) Is Luzia a net creditor or net debtor? Compute its net external wealth W in lúz.

$W = \text{Assets} - \text{Liabilities} = 400 - 600 = -200$ billion lúz. Luzia is a **net debtor**.

- (b) Given the share of the external wealth of Luzia in USD, calculate the net dollar-denominated position of Luzia. Is Luzia a net borrower or lender of USD denominated assets?

External assets are entirely in luz (no dollar exposure). Dollar-denominated liabilities: $0.80 \times 600 = 480$ billion luz = 480 billion dollars (at $E = 1$).

Net dollar exposure = $0 - 480 = -480$ billion dollars. Luzia is *net short* in dollars: it has borrowed 480 billion dollars without holding any dollar assets to offset the exposure.

- (c) The luz depreciates to $E = 1.25$ luz per dollar (a 25% depreciation). Compute the *valuation effect* — the change in W due solely to the exchange rate change — measured in luz. Does Luzia become richer or poorer?

After depreciation, dollar-denominated liabilities now cost $1.25 \times 480 = 600$ billion luz (up from 480). Assets are unchanged in luz. New $W = 400 - (120 + 600) = 400 - 720 = -320$ billion luz.

Valuation effect = $-320 - (-200) = -120$ billion luz. Luzia becomes **poorer**: the depreciation inflates the luz value of its dollar-denominated debt while leaving its luz assets unchanged. This is a standard *original-sin* problem — borrowing in foreign currency exposes the country to adverse valuation effects when the domestic currency weakens.

3. (12 points) Fiscal Dominance and Reserve Depletion.

A country maintains a fixed exchange rate. Its money supply is currently 3,000 million pesos, domestic credit is 1,500 million pesos, and reserves are 1,500 million pesos. Each year the government runs a deficit that the central bank monetizes by buying 400 million pesos of government bonds. Prices are sticky and output is unchanged.

- (a) Fill in the table below, showing domestic credit, reserves, and the backing ratio for years 1 through 4. (Note: the money supply stays constant as long as the peg holds and P , Y are fixed.)

Each year: B rises by 400, R falls by 400, M stays at 3,000.

Year	M	B	R	Backing Ratio	Peg status
1	3,000	1,500	1,500	50%	Holds
2	3,000	1,900	1,100	36.7%	Holds
3	3,000	2,300	700	23.3%	Holds
4	3,000	2,700	300	10%	Holds
3.75 [†]	3,000	3,000	0	0%	Breaks

[†] Peg collapses partway through year 4 when reserves reach 0. Starting from 300 at the beginning of year 4, reserves are exhausted after $300/400 = 0.75$ years.

- (b) Suppose investors are forward-looking and fully anticipate the reserve trajectory. What happens before reserves actually reach zero? Explain the logic of a speculative attack in this

context.

Rational investors know that once the currency floats, the money supply will grow at the rate of domestic credit expansion, causing inflation and depreciation. To avoid holding an asset whose value is about to fall, investors sell domestic currency *before* reserves are exhausted. This triggers a sudden, large demand for foreign currency (a speculative attack) that drains the remaining reserves at once — collapsing the peg *earlier* than the myopic calculation suggests.

The attack occurs when the shadow floating exchange rate (the rate that would prevail immediately after floating) equals the peg rate. At that point, there is no expected capital gain from attacking, but also no loss — so rational investors attack immediately. In practice, the peg collapses well before year 4.

4. (12 points) Non-Credible Peg and Self-Fulfilling Crisis.

Country Sola pegs its currency (the sol) to the dollar at a 1-to-1 rate (1 sol = 1 USD). The US interest rate is $i^* = 3\%$. Initially the peg is credible: expected depreciation is zero.

(a) Under a credible peg, what is the sol interest rate? Show this using the UIP condition.

With a credible peg, $E^e = E$ (expected depreciation = 0), so UIP gives:

$$i_{\text{sol}} = i^* + 0 = 3\%.$$

The sol rate equals the dollar rate.

(b) Suppose investors start to doubt the peg and assign a 20% probability that the sol will depreciate by 15% against the dollar over the next year. Compute the expected depreciation term and the new equilibrium sol interest rate (assume the peg still holds in the short run).

Hint: to calculate the expected depreciation you can multiply the possible depreciations times their probabilities and sum. This is known as finding an *expected value*.

Expected depreciation:

$$\Delta^e = 0.20 \times 0.15 + 0.80 \times 0 = 0.03 \quad (3\%).$$

UIP now requires:

$$i_{\text{sol}} = 3\% + 3\% = 6\%.$$

The central bank must raise the interest rate by 3 percentage points (by contracting the money supply, shifting LM left) to maintain the peg. This contraction reduces output — exactly what makes a non-credible peg more costly to defend than a credible one.

(c) Explain, in one paragraph, how a loss of credibility can become a *self-fulfilling crisis*, even

without any deterioration in economic fundamentals.

If investors believe the peg will break, they demand a higher interest rate (currency premium). The central bank must contract the money supply to deliver this rate, which depresses output. Lower output makes the government's commitment to the peg — and the associated output cost — politically untenable, increasing the likelihood that it will actually abandon the peg. This validates the investors' initial belief. The crisis is self-fulfilling: the *expectation* of depreciation forces the policy tightening that causes the recession that causes the peg to break, even if none of this would have happened had investors kept believing in the peg. This is the “Zone II” equilibrium multiplicity from second-generation crisis models.

5. (10 points) PPP, Fisher Effect, and UIP.

Country Aqua has annual inflation $\pi_A = 10\%$; Country Bora has $\pi_B = 2\%$. The current spot rate is $E_{A/B} = 5$ (Aqua per Bora).

- (a) According to relative PPP, what is the expected rate of depreciation of the Aqua currency against the Bora currency? Over what horizon is this prediction most reliable?

Relative PPP predicts:

$$\hat{E}_{A/B}^e \approx \pi_A - \pi_B = 10\% - 2\% = 8\%.$$

The Aqua currency is expected to depreciate by 8% per year against the Bora. This prediction is most reliable in the *long run* — short-run deviations from PPP are common due to sticky prices, demand shocks, and financial frictions.

- (b) If the Bora nominal interest rate is $i_B = 4\%$, what does the Fisher effect predict for the Aqua nominal interest rate?

Hint: You can use the equalization of real rates (Real Interest Rate Parity) together with the Fisher equation, or the UIP-PPP to find the answer.

The Fisher effect: $i \approx r + \pi^e$. If the real interest rate is equalized internationally (real interest parity), then:

$$i_A \approx i_B + (\pi_A - \pi_B) = 4\% + 8\% = 12\%.$$

Alternatively, this follows directly from combining relative PPP with UIP: $i_A = i_B + \hat{E}^e = 4\% + 8\% = 12\%$.

- (c) Using UIP, confirm that the nominal interest differential is consistent with the depreciation found in (a).

UIP requires $i_A = i_B + (E_{A/B}^e - E_{A/B})/E_{A/B}$. We have $i_A - i_B = 12\% - 4\% = 8\%$, and the depreciation term from (a) is also 8%. They are consistent:

$$12\% = 4\% + 8\%. \quad \checkmark$$

This consistency is not a coincidence — combining relative PPP (which pins the depreciation) with the Fisher effect (which links inflation to nominal rates) directly implies UIP holds at the predicted rates.

6. (12 points) Speculative Attacks.

- (a) What is a speculative attack? Describe the sequence of events in terms of investor behavior, reserve depletion, and peg collapse.

A speculative attack occurs when investors simultaneously convert large holdings of domestic currency into foreign currency, expecting the peg to break. The sequence: (1) Investors expect the domestic currency to depreciate; (2) they sell domestic assets and buy foreign currency from the central bank; (3) the central bank's reserves fall rapidly; (4) once reserves are exhausted, the central bank can no longer defend the peg and is forced to float; (5) the currency depreciates, validating the investors' expectation. Attacks can be self-fulfilling: even a country with sound fundamentals can be attacked if investors believe an attack is coming, because the central bank's response (contracting the money supply) is itself output-costly and politically unsustainable.

- (b) Consider a country with fundamentals that are currently sound but deteriorating (growing domestic credit, shrinking reserves). Explain why rational investors attack *before* reserves are actually zero, rather than waiting.

If investors wait until reserves reach zero, the currency floats at a *lower* value than the peg — meaning investors who held domestic currency until the last moment suffer an immediate capital loss at the moment of collapse. Rational investors therefore attack earlier — specifically when the “shadow” exchange rate (the rate that would prevail if the currency were floating right now) equals the peg rate. At that point, attacking causes no further loss, and waiting causes a certain loss. The incentive to attack is thus already present well before reserves are zero. The attack brings the collapse forward, and it happens in a sudden rush because all investors face the same incentive simultaneously.

- (c) What are two policy tools a government can use to reduce the likelihood of a speculative attack, and what is the main drawback of each?

(1) **Accumulate reserves:** a larger reserve buffer makes it more costly to exhaust reserves, reducing attack incentives. Drawback: accumulating reserves requires sustained current

account surpluses or capital inflows, which is not always feasible; holding reserves also has an opportunity cost.

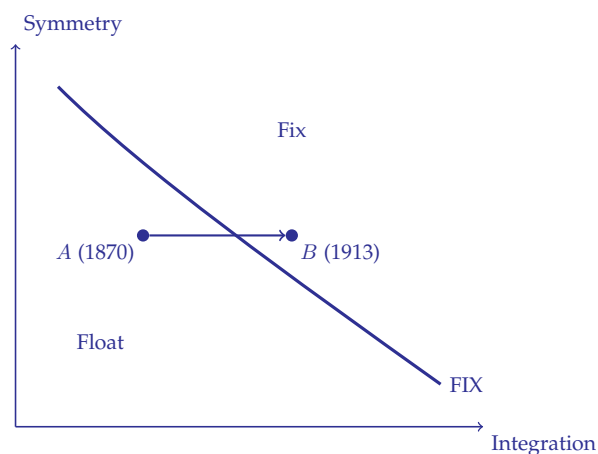
(2) **Capital controls:** restricting the ability of investors to sell domestic currency prevents the coordinated run. Drawback: capital controls create distortions, may signal weakness (spooking investors further), can be evaded through financial innovation, and reduce the country's ability to attract foreign investment over the long run.

7. (12 points) OCA – Gold Standard and Symmetry.

Between 1870 and 1913 world trade relative to GDP roughly doubled. Between 1913 and 1939 it fell back to approximately its 1870 level, partly due to rising tariffs and quotas.

- (a) Using the symmetry-integration diagram, explain why the rise in trade between 1870 and 1913 tended to support the gold standard (a form of multilateral fixed exchange rate system). Sketch the relevant shift.

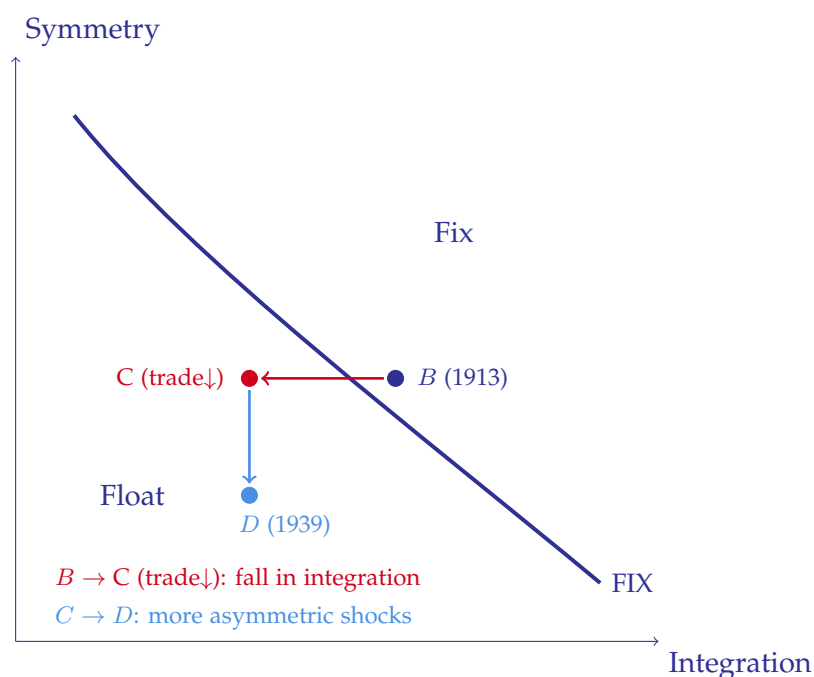
Higher trade increases economic integration (the horizontal axis in the diagram). Holding symmetry roughly constant, the point representing a typical country pair shifts to the right. If it crosses above the FIX line, the net benefit of fixing becomes positive. The gold standard was thus self-reinforcing: trade growth made fixing more attractive, and fixing (by reducing currency uncertainty) further boosted trade. This is an example of *endogenous OCA*: the criteria become satisfied by the act of fixing.



Moving from *A* to *B* (rightward along integration axis), the point crosses the FIX line, making the gold standard optimal.

- (b) Between 1913 and 1939, economists have noted an increase in country-specific shocks —i.e., individual, for example effects from the war, depression— alongside the (global) collapse in trade. Using the diagram, show *two separate* forces that undermined the gold standard.

Two forces shift the equilibrium unfavorably: (1) The *fall in trade* reduces integration, shifting the point left (lower horizontal axis value). (2) The *increase in country-specific shocks* reduces shock symmetry, shifting the point downward (lower vertical axis value). Both forces push the point below the FIX line, making the gold standard suboptimal.

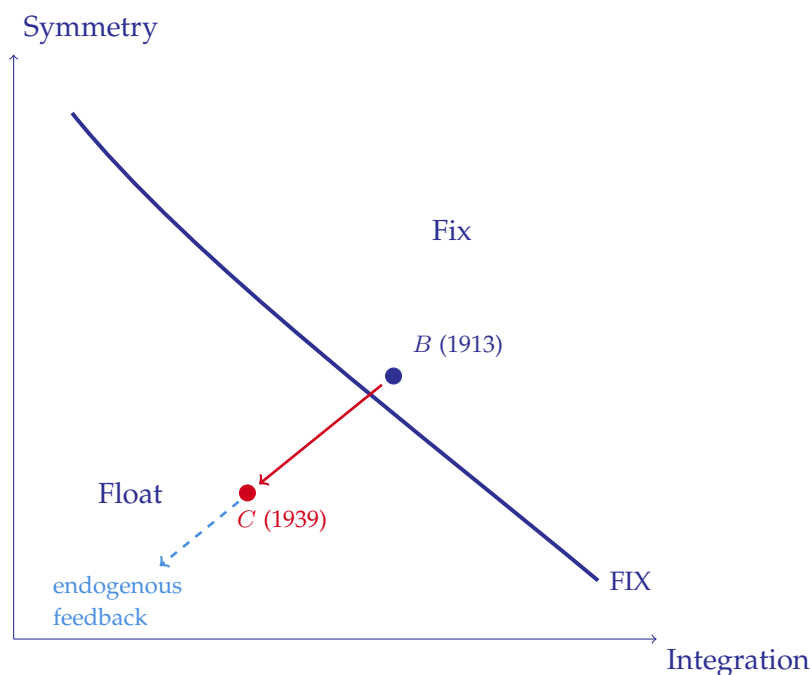


Starting from *B* (1913, above FIX), the trade collapse shifts the point left (red arrow leading to *C*) and the rise in country-specific shocks shifts it further downward (blue arrow), landing at *D* (1939) well below the FIX line. Both forces independently push the pair into the Float region, making the gold standard suboptimal.

- (c) How does the collapse of the gold standard in the 1930s illustrate the concept that OCA criteria can be *endogenous*?

When countries left the gold standard and trade barriers rose, bilateral trade fell — which in turn reduced integration and made fixing less attractive, inducing more countries to exit. The causality runs in both directions: lower integration causes floating, and floating reduces integration further. Similarly, asymmetric shocks caused exit, and exit allowed idiosyncratic monetary policies that generated further divergence. OCA criteria are therefore partly endogenous to the exchange rate regime itself.

The diagram illustrates this feedback loop: moving from *B* to *C* was not just an exogenous shift — leaving the gold standard itself contributed to the further leftward and downward drift, as shown by the reinforcing arrows below.



The dashed blue arrow represents the endogenous feedback: once at C , floating itself further erodes integration and symmetry, pushing the point even further from the FIX line. The initial exogenous shocks (war, depression, trade collapse) triggered the exit, but the exit then amplified the very criteria that made re-entry less attractive — a self-reinforcing dynamic away from the gold standard.

8. (10 points) Hyperinflation and the Fixed Exchange Rate as a Nominal Anchor.

Country Inflancia has experienced several years of hyperinflation due to persistent fiscal deficits financed by money printing. Its central bank is considering pegging the currency to the US dollar.

- (a) Explain why adopting a credible fixed exchange rate (a currency peg) can reduce inflation. Use the concepts of fiscal discipline and inflation expectations in your answer.

A credible peg to the dollar forces the domestic interest rate to equal the US rate (by UIP with zero expected depreciation). This means the central bank can no longer print money to finance deficits without losing reserves and breaking the peg. The political cost of peg abandonment creates implicit fiscal discipline: the government is constrained from excessive money financing.

On the inflation front, we can see how by the Fisher equation and the real interest rate parity, equal nominal rates implies that the home economy will stabilize its inflation to that of the foreign economy:

$$r = r^*$$

$$i - \pi = i^* - \pi^*$$

$$\dot{x} - \pi = \dot{x}^* - \pi^*$$

On the expectations side, if the peg is credible, agents expect low inflation (consistent with the US inflation rate), which reduces nominal wage and price-setting pressure. Lower expected inflation reduces realized inflation through wage contracts and price indexation. Together, fiscal discipline and anchored expectations can break the hyperinflationary spiral without requiring as large an output sacrifice as would be needed under floating.

(b) What determines whether the peg will actually be credible? List two factors.

(1) **Adequate reserves:** the central bank must hold sufficient foreign reserves to withstand speculative attacks and defend the peg under adverse conditions. A backing ratio close to 100% (as in a currency board) maximizes credibility.

(2) **Fiscal sustainability:** if the government continues running deficits that require money financing (fiscal dominance), the central bank will eventually deplete reserves. Investors know this and will not view the peg as permanent. A credible commitment to fiscal adjustment is therefore essential for peg credibility.

Among other possible factors, we can include anything that implies an important money emission (increases in money supply), e.g., whether the economy is approaching a recession, or a if a war is underway, etc. These type of factors will increase the expectations of ER depreciations.

(c) Describe one situation in which the costs of a fixed exchange rate as a nominal anchor could outweigh the benefits.

If the country suffers a large *asymmetric recession* while the anchor country (US) is booming, the anchor country's optimal monetary policy (tight, to cool the boom) is the opposite of what the country needs (loose, to fight the recession). By pegging, the country must raise its interest rate to match the US, deepening its own recession. The output cost of the peg can become politically and economically intolerable, ultimately forcing abandonment — as occurred with Argentina's currency board in 2001.

9. (20 points) Speculative Attacks and Peg Collapse.

A small open economy has maintained a fixed exchange rate $\bar{E}$ for several years, but persistent current account deficits have steadily eroded its stock of foreign reserves $\mathcal{R}$. Market participants begin to doubt that the central bank has sufficient reserves to sustain the peg. As a result, they revise upward their expected future exchange rate: $E_{h/f}^e \uparrow$. The central bank attempts to defend $\bar{E}$ by contracting the money supply, but reserve losses accelerate as the attack intensifies.

(a) Recall that the foreign return in the FX market is:

$$FR = i^* + \frac{E_{h/f}^e - E_{h/f}}{E_{h/f}}$$

and that under uncovered interest parity (UIP), the domestic interest rate must satisfy:

$$i = i^* + \frac{E_{h/f}^e - E_{h/f}}{E_{h/f}}$$

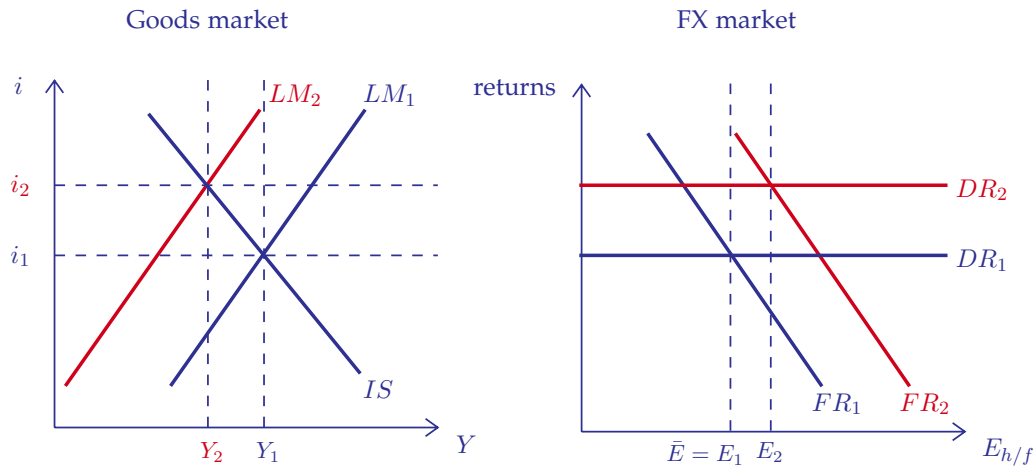
How does the revision in expectations ($E_{h/f}^e \uparrow$) affect the FR curve? What interest rate must the central bank maintain to keep $E_{h/f} = \bar{E}$, and what does this imply for monetary policy and output? Use an IS-LM-FX diagram to illustrate.

When $E_{h/f}^e \uparrow$, foreign assets become more attractive at every (spot) exchange rate: the FR curve shifts **up**, creating depreciation pressure on the home currency.

To keep $E_{h/f} = \bar{E}$, UIP requires:

$$i = i^* + \frac{E_{h/f}^e - \bar{E}}{\bar{E}}$$

Since $E_{h/f}^e > \bar{E}$, the required domestic interest rate is **above** i^* . The central bank must raise i by selling foreign reserves and buying domestic currency, contracting the money supply ($M \downarrow$) and shifting LM **left**. With IS unchanged (the exchange rate has not actually moved, so there is no competitiveness effect), the new goods-market equilibrium has lower output and a higher interest rate.



Effects of the defense phase:

- FR shifts right/up: depreciation pressure at $\bar{E}$
- IS unchanged: exchange rate has not moved
- LM shifts left: CB contracts money supply to raise i

- $Y \downarrow, i \uparrow, E_{h/f} = \bar{E}$ (peg holds, for now)
- $\mathcal{R} \downarrow$: foreign reserves falling

Key insight: The stronger the loss of credibility (the larger $E_{h/f}^e - \bar{E}$), the higher the interest rate required and the larger the output contraction needed to defend the peg.

- (b) Describe the dynamic of a speculative attack. If private agents believe the peg will collapse, what do they do, and how does this affect the central bank's reserve position? Why is the attack self-fulfilling?

If agents believe the peg will collapse and the currency will depreciate, they have a strong incentive to attack immediately:

- They sell home-currency assets and buy foreign-currency assets (capital flight).
- To defend $\bar{E}$, the CB must absorb all of this selling, buying home currency with its foreign reserves.
- This drains $\mathcal{R}$ rapidly.

The self-fulfilling nature arises because the attack itself causes the reserve depletion that makes collapse inevitable:

- Even if the CB *could* have sustained the peg absent the attack, the sudden surge in reserve demand forces collapse.
- Agents who attack profit if the peg collapses: they bought foreign currency cheaply at $\bar{E}$ and now hold it at the post-collapse depreciated rate.
- The dominant strategy for any agent who believes others will attack is therefore to attack as well.

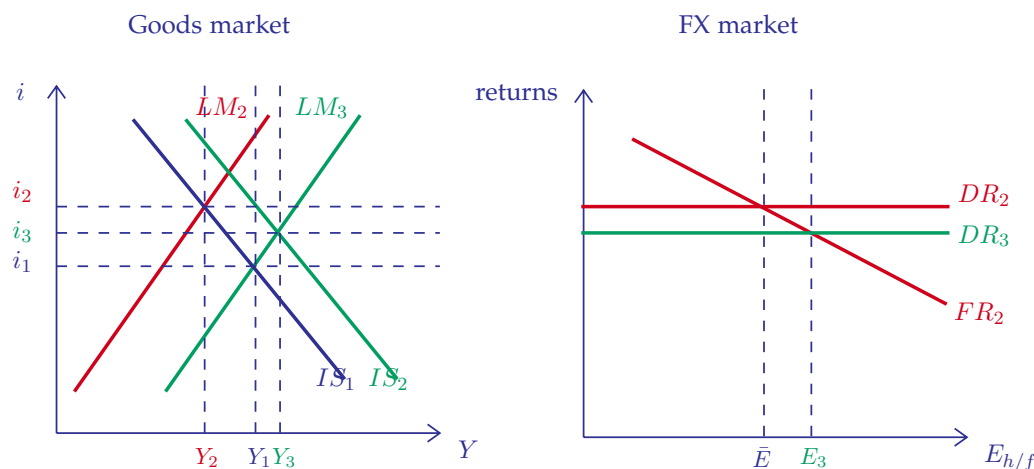
This coordination problem means that even a fundamentally sound peg can collapse if credibility is lost. It is a key insight from these models (called second-generation currency crisis setups): a peg may be in a *bad equilibrium* simply because everyone expects it to be.

- (c) Once reserves are exhausted the central bank abandons the peg. Using IS-LM-FX diagrams, trace the effects of the collapse. What happens to Y , i , and $E_{h/f}$ relative to (i) the pre-crisis equilibrium and (ii) the defense phase? Summarize the full sequence in a table.

When the peg is abandoned, two things happen simultaneously:

- The CB releases the money supply constraint: LM shifts back right toward its pre-crisis position.
- The exchange rate jumps to its new market-clearing level $E_3 > \bar{E}$ (large depreciation).

This improves competitiveness and the trade balance: IS shifts right.



Note: In this plot we assume LM, and DR go back to their previous (pre-shock) positions, that is: $LM = LM_3$, $DR = DR_3$. This is why the LM, DR curves are not visible (the new ones are superimposed on them).

Summary of the full sequence:

	Y	i	$E_{h/f}$
Pre-crisis equilibrium	Y_1	i_1	$\bar{E}$
Defense phase	$Y_2 < Y_1$	$i_2 \gg i_1$	$\bar{E}$
Post-collapse equilibrium	$Y_3 > Y_2$	$i_1 < i_3 < i_2$	$E_3 > \bar{E}$

The economy endures a sharp contraction during the defense phase, then partially recovers after the collapse as the depreciation boosts competitiveness and the money supply constraint is lifted. Whether Y_3 exceeds Y_1 is ambiguous: it depends on the relative magnitude of the rightward IS shift and the extent to which LM recovers. This boom-bust pattern is a characteristic feature of historical currency crises.

- (d) The boom-bust pattern in the table above is a stylized feature of historical currency crises. Briefly identify one real-world episode that fits this pattern and explain which elements of the model are most consistent with what happened, and where the model falls short.

Example: East Asian Crisis, 1997–98 (Thailand, Indonesia, South Korea).

These countries maintained de facto pegs to the U.S. dollar. When doubts arose about reserve adequacy and current account sustainability, investors attacked the pegs. Central banks raised interest rates sharply to defend — consistent with the defense phase: $i \uparrow\uparrow$, $Y \downarrow\downarrow$, reserves draining rapidly. Once pegs collapsed, currencies depreciated 30–50%, export competitiveness recovered, and output began recovering within 1–2 years — consistent with the post-collapse IS shift and recovery toward Y_3 .

Most consistent model elements:

- The self-fulfilling nature of the attack: Thai fundamentals were weak but not obviously crisis-inducing absent the confidence shock.
- The contractionary defense phase: Thai interest rates exceeded 20% annualized during the defense period.
- The post-collapse depreciation-driven recovery: Korean exports surged after the won depreciated, shifting IS rightward as the model predicts.

Where the model falls short: Balance-sheet effects. Many firms and banks had dollar-denominated liabilities. The depreciation that the model predicts should be expansionary (IS right) was partially offset by balance-sheet contractions, as the domestic-currency value of foreign debt surged. This channel is absent from the basic IS-LM-FX framework and helps explain why the post-collapse recovery was slower and more painful than the model suggests.